

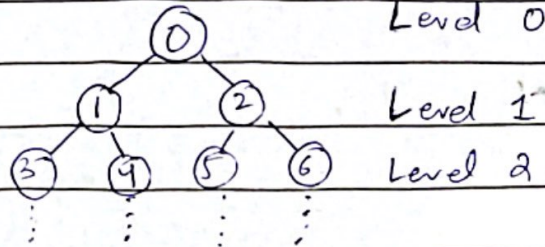
BINARY TREES Properties

Date

M	T	W	T	F	S	S
---	---	---	---	---	---	---

Syed M. Asad Ali (sa09475).

(i) The maximum number of nodes at level, L of a binary tree is 2^L .



Level 0

Level 1

Level 2

$$L_0 = 2^0 = 1$$

$$L_1 = 2^1 = 2$$

$$L_2 = 2^2 = 4$$

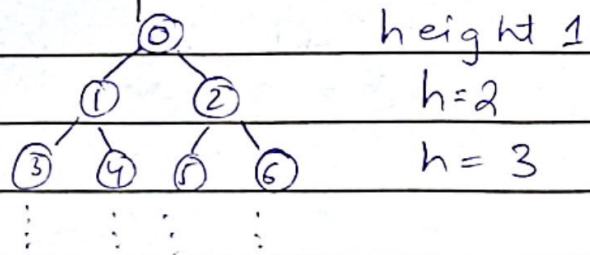
$$L_3 = 2^3 = 8$$

$$\vdots$$

$$L_i = 2^i = 2^i$$

Hence, maximum number of nodes at level $\{L\}$ is 2^L .

(ii) The maximum number of nodes in a Binary Tree of height, h i.e. the total number of nodes in tree is; $2^h - 1$.



height 1

$h=2$

$h=3$

$$H_1 = 1 = 2^1 - 1$$

$$H_2 = 3 = 2^2 - 1$$

$$H_3 = 7 = 2^3 - 1$$

$$H_4 = 15 = 2^4 - 1$$

$$\vdots$$

$$H_i = 2^i - 1$$

→ Hence, then maximum number of nodes of binary tree of height $\{h\}$ is $2^h - 1$.

(iii) In a Binary Tree with N nodes, minimum possible height or minimum number of levels is $\log_2(N+1)$.

Page No. _____

RC

Signature _____

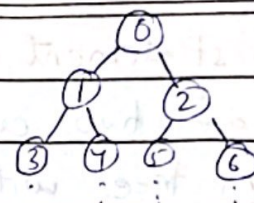
Date _____

M	T	W	T	F	S	S
---	---	---	---	---	---	---

$$N = 1, h = 1,$$

$$N = 3, h = 2$$

$$N = 7, h = 3$$



N = 1

N = 3

N = 7

$$N = N, h = h$$

$$\rightarrow N = 2^h - 1 \Rightarrow N + 1 = 2^h \Rightarrow \boxed{\log_2(N + 1) = h}$$

$\rightarrow$ Hence, a binary tree with N nodes has minimum of $\log_2(N + 1)$.

(iv) A binary tree with n_L leaves has at least $\text{ceil}(\log_2 n_L) + 1$ levels.

$\rightarrow$ For a binary tree with L levels, the last level contains at most 2^L nodes. Therefore,

$$n_L \leq 2^L \Rightarrow L \geq \log_2(n_L)$$

since L is an integer, we round off $\log_2(n_L)$, round up in this case as a level with even 1 node is considered new level.

$$L \geq \text{ceil}(\log_2(n_L))$$

we add 1 for the root node:

$$L \geq \text{ceil}(\log_2(n_L)) + 1$$

(v) In a Binary Tree where each node has 0 or 2 children, the number of leaf nodes is always one more than nodes with two children, i.e., $n_L = n_i + 1$.

Base case: ($n_i = 0$) root node or a case where there are no node with two children. $n_L = n_i + 1$

$$n_L = n_i + 1$$

$$n_L = 0 + 1$$

$$n_L = 1$$

$\Rightarrow$ The statement holds true for n_i nodes.

Page No. _____

RC

Signature _____

⇒ The statement also holds true for $n_i + 1$ nodes.

⇒ There are two cases for new node added to already existing tree with " n_i " nodes with two children.

(i) when node has no children it is a leaf node.

In this case leaf nodes increase by 1.

$$n_L = (n_i + 1) + 1$$

$$n_L = n_i + 2$$

(ii) when node has two children. In this case no. of nodes with two children increase by 1.

$$n_L = (n_i + 1) + 1$$

$$n_L = n_i + 2$$

(vi) If a binary tree has n_i internal nodes then total no. of nodes

$$N = 2n_i + 1$$

Total no. of internal nodes = n_i

Total no. of leaf nodes = n_L

Total no. of nodes = $n_i + n_L$

From (v) we know $n_L = n_i + 1$, Hence;

$$N = n_i + n_i + 1$$

$$N = 2n_i + 1$$

(vii) If a binary tree has a total of N nodes, then no. of internal nodes is :

$$n_i = (N - 1) / 2$$

From (vi) we know $N = 2n_i + 1$, Hence;

$$N - 1 = 2n_i \Rightarrow n_i = (N - 1) / 2$$

(viii) If a binary tree has a total of N nodes, the no. of leaves is :

$$n_L = (N + 1) / 2$$

Total no. of leaf nodes = n_i

Total no. of internal nodes = n_L

Total no. of nodes = N

→ The leaf nodes are as many as the difference of no. of all nodes and no. of all internal nodes: $n_L = N - n_i$

→ From (vii) we know $n_i = \frac{N-1}{2}$, hence,

$$n_L = N - \frac{N-1}{2}$$

$$n_L = \frac{2N - N + 1}{2} \Rightarrow \boxed{n_L = \frac{N+1}{2}}$$

(ix) If a tree has n_L leaves, then total no. of nodes is:

$$N = 2n_L - 1$$

From (viii) we know $n_L = \frac{N+1}{2}$, hence,

$$2n_L = N+1 \Rightarrow \boxed{N = 2n_L - 1}$$

(x) If a Binary tree has n_L leaves, then no. of internal node is: $n_i = n_L - 1$

From (vi) and (ix) we know that:

$$N = 2n_i + 1 \text{ and } N = 2n_L - 1$$

$$\Rightarrow 2n_i + 1 = 2n_L - 1$$

$$\Rightarrow 2n_i = 2n_L - 1 - 1$$

$$\Rightarrow 2n_i = 2n_L - 2$$

$$\Rightarrow 2(n_i) = 2(n_L - 1) \Rightarrow \boxed{n_i = n_L - 1}$$